Luiz Braga

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Software developer with a Bachelor's degree in Physics and hands-on experience in backend development and data engineering. Skilled to manage ETL pipelines, microservices, and real-time data processing systems. Proficient in Python, JavaScript, Rust, and SQL, with experience working with databases like PostgreSQL, BigQuery, and Neo4j. Experienced in deploying containerized applications using Docker, data extraction and transformation, and integrating messaging systems. Adept at solving complex problems and learning new technologies as needed to deliver **robust**, **optimal** and **reusable** software.

Technical Skills

- **Programming Languages:** Python, JavaScript, Typescript, Go, Rust, Julia, SQL, GraphQL, Bash, Modern Fortran and C
- **Databases:** PostgreSQL, BigQuery, Neo4j
- **Tools & Frameworks:** Docker, ElasticSearch,, GraphQL, GraphAPIs, Linux API, Numpy, Scikit-learn
- **Data Engineering:** ETL pipelines, data extraction, cleansing, transformation, web scraping, distributed systems
- **Software Development:** Microservices, containerization, optimization, bytecode compilation, memory management and data oriented design
- **Other Skills:** regex, scientific programming, linear algebra and calculus

Professional Experience

Backend Developer
Notus Labs (2023–2024)

- Designed a system where a **Rust-based server** received HTTP requests for arbitration, fetched real-time market data from **CoinGecko API**, and communicated with a **Julia backend** via **Linux sockets**.
- Implemented a **Julia service** to parse market data, perform numerical minimization calculations, and determine optimal arbitration opportunities.
- Enabled seamless communication between Rust and Julia using **Linux sockets**, ensuring efficient data transfer and low-latency processing.
- Optimized the arbitration algorithm to handle large data volumes, improving execution speed and computational efficiency.
- Delivered a robust solution for identifying and exploiting market inefficiencies in real-time.
- Improved arbitration algorithms for the crypto asset market, enhancing computational efficiency and execution speed.
- Wrote **SQL queries** for fast and accurate data extraction from multiple sources, supporting backend financial services.
- Built interactive dashboards for real-time company data monitoring, aiding decision-making processes.

Data Engineer
Heimdall (2022–2023)

- Developed **Python scripts** using **Beautiful Soup** (bs4) and **Bash scripts** to extract and parse transaction data from blockchain explorers.
- Collected and processed large volumes of transaction data to simulate real-world scenarios for testing security algorithms.
- Validated and cleansed scraped data to ensure accuracy and reliability for analysis.
- Supported the development and refinement of blockchain security algorithms by providing realistic datasets for testing fraud detection and transaction integrity.
- Automated data collection workflows to enable continuous updates and testing of security models.
- Researched and implemented algorithms for **blockchain security** and data integrity, contributing to secure transactions and fraud prevention.
- Managed relational (**PostgreSQL**) and non-relational databases (**Neo4j**, **BigQuery**) to handle structured and unstructured data.
- Migrated **BigQuery** databases for asset sale.
- Developed **GraphAPIs** using **GraphQL** for flexible and efficient data queries.
- Maintained ETL pipelines using Python, SQL, and Neo4j to collect, transform, and load social media data

from Twitter.
- Manage a system to analyze user sentiment and predict cryptocurrency market trends using **Python** and **Scikit-learn**.
- Processed large volumes of social media data to identify patterns and correlations between user sentiment and crypto asset performance.
- Utilized **Neo4j** to model and query complex relationships in the data, enabling insights into market behavior.
- Integrated machine learning models to classify sentiment and forecast potential market movements.
- Automated data collection and transformation workflows, ensuring timely and accurate updates for analysis.
- Developed and deployed **data analyzers** and **crawlers** as microservices, running on **Google Cloud**.
- Utilized **Docker** to containerize each service, ensuring consistency across development, testing, and production environments.
- Orchestrated microservices using **Docker Compose**, simplifying deployment and management of multi-container applications.
- Enabled scalable data processing by leveraging cloud infrastructure.
- Improved system reliability and maintainability by isolating services and automating deployment workflows.
- Monitored microservices using **ElasticSearch** to ensure system reliability and performance.

Relevant Projects

Bytecode Compiler - Lolilang

- Created a bytecode compiler to process and evaluate text commands
- Implement a garbage collector for **automatic memory management**.
- Implemented features like **variable declarations**, **functions**, **closures**, **arrays**, **hash maps**, and control structures such as **conditionals** and **loops**.
- Added support for **mathematical operations**, including recursion, and studied memory management to improve code efficiency.
- Implement built-in function for HTTP requests, complex number evaluation and frame manipulation.

Terminal-Based Text Editor - Nino

- Developed a lightweight, terminal-based text editor using the **Linux API**, inspired by **Vim**.
- Added **syntax highlighting** and advanced search functionality using **regular expressions (regex)** for efficient text searching.
- Focused on **efficiency** and **simplicity**, providing a streamlined experience for developers on Unix-like systems.

Scientific Programming Video Lessons in Modern Fortran - YouTube

- Produced and published a series of video lessons on **scientific programming** using **Modern Fortran**.
- Covered topics including **data structures**, **I/O**, **OOP**, **polymorphism**, **matrices**, and **numerical programming**.
- Focused on practical examples and theoretical concepts for scientific and engineering applications.

Education

- **Federal University of Santa Catarina** — Bachelor’s Degree in Physics